

Candidate Name

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

CHEMISTRY
PAPER 5 Practical Test

9189/5**NOVEMBER 2015 SESSION**

1 hour 20 minutes

Candidates answer on the question paper.

Additional materials:

As listed in Instructions to Supervisors

Electronic calculator

TIME 1 hour 20 minutes**INSTRUCTIONS TO CANDIDATES**

Write your name, Centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

You are advised to show all working in calculations.

Use of a Data Booklet is unnecessary.

FOR EXAMINER'S USE

1	
2	
TOTAL	

This question paper consists of 7 printed pages and 1 blank page.

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- 1 You are required to estimate the percentage composition of iron (II) ions in a brown contraceptive pill. The major component of the pill is, ferrous fumarate, $\text{C}_4\text{H}_2\text{FeO}_4$.

For
Examiner's
Use

FA1 are three contraceptive pills.

FA2 is a $0.001 \text{ mol dm}^{-3} \text{ KMnO}_4 (\text{aq})$.

- (a) Weigh an empty small beaker.

Place **FA1** (the three pills) in the small beaker and weigh again.

Record your weighings in **Table 1.1**.

Table 1.1

mass of small beaker + FA1 /g	
mass of empty small beaker / g	
mass of FA1 used /g	

[2]

Crush the three pills using mortar and pestle.

Transfer all the powder into the small beaker.

Using a measuring cylinder add 15 cm^3 of hot 1.0 mol dm^{-3} sulphuric acid to the crushed pills in the small beaker and stir vigorously with a glass rod.

Transfer all the contents of the beaker into a 250 cm^3 volumetric flask. Make up to the mark with distilled water.

Mix the contents thoroughly by shaking and label this solution **FA3**.

- (b) Filter **FA3** into a clean beaker

Pipette 25.00 cm^3 of the filtered **FA3** into a conical flask and add 10.00 cm^3 of 1.0 mol dm^{-3} sulphuric acid.

Titrate the contents of the conical flask with **FA2**.

Repeat the titration as many times as you think necessary to obtain accurate results.

Record your burette readings in Table 1.2.

Table 1.2:

final burette reading / cm ³				
initial burette reading / cm ³				
volume of FA2 used / cm ³				

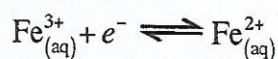
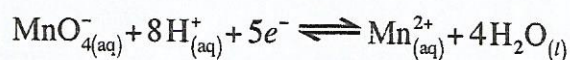
[15]

Summary

25.00 cm³ of FA3 reacted with _____ cm³ of FA2.

Show which results you used to obtain this volume of FA2 by placing a tick under the readings in Table 1.2.

In acidic conditions, MnO₄⁻ and Fe²⁺ ions react according to the following half equations:



- (c) Write a balanced equation for the reaction between Fe²⁺ and MnO₄⁻.

[1]

- (d) Calculate the number of moles of MnO₄⁻ in the volume of FA2 which reacted with 25 cm³ of FA3.

[1]

- (e) Calculate the number of moles of $\text{Fe}_{(\text{aq})}^{2+}$ in 25.00 cm^3 of FA3.

[1]

- (f) Determine the mass of Fe^{2+} in the contraceptive pill.

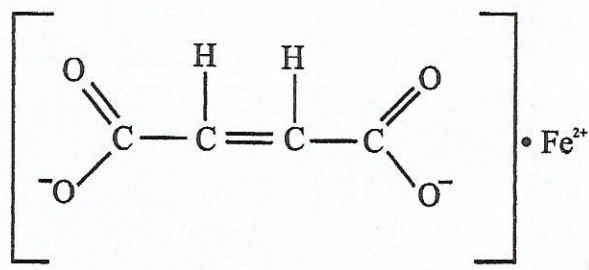
$[A_r(\text{Fe}): 55.8]$

[2]

- (g) Estimate the percentage composition of iron in the contraceptive pill.

[1]

- (h) Given that the structural formulae of ferrous fumarate is



suggest why the value of the percentage composition of iron determined in the experiment is much higher than the actual value.

[1]

- (i) Describe and explain how the end point is observed in this analysis.

[2]

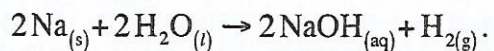
[Total :26]

ASSESSMENT OF PLANNING SKILLS

DO NOT CARRY OUT YOUR PLAN.

- 2 Sodium is a very reactive metal which is stored under oil to prevent contact with air and water vapour.

The reaction of sodium with water can be represented by the equation,



You are required to determine the A_r of sodium using a given sample of sodium metal basing on the above reaction.

Present your plan as a sequence of numbered steps and draw a diagram showing the set up of the experiment. Your plan should show how you would use your results to determine the A_r of sodium.

Assume you are provided with the following:

a piece of sodium metal
water
thistle funnel
absorbent paper
rubber stopper
delivery tube
measuring cylinder
beaker

[molar gas volume, V_m , = $24 \text{ dm}^3 \text{ mol}^{-1}$ at r.t.p.]

The plan

[Total :14]